NNN-C-25B
July 10, 1978
SUPERSEDING
Int. Fed. Spec. NNN-C-0025A(DSA-DM)
May 20, 1975 and
Fed. Spec. NNN-C-25
March 19, 1963

FEDERAL SPECIFICATION

CASSEROLE, LABORATORY, Porcelain

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers porcelain casseroles suitable for general purpose use.

1.2 Classification.

1.2.1 Types, classes and capacities. Porcelain casseroles covered by this specification shall be of the following types, classes, and capacities, as specified (see 6.2):

Type I - Porcelain cup, porcelain handle, no cover.

Class 1 - General laboratory use
Capacity, milliliters (ml)
30
140
210
340
500
750

Class 2 - Silica determination
300 ml

Type II - Porcelain cup, wooden handle, porcelain cover.
150 ml (see 6.3)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

FSC 6640

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Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup, Paperboard.

Federal Standard:

FED-STD-123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards, required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Society for Testing and Materials (ASTM) Standards:

C242 - Terms Relating to Ceramic Whiteware and Related Products.
C327 - Standard Method of Test for Linear Thermal Expansion of Fired Ceramic Whiteware Materials by the Interferometric Method.
C372 - Standard Method of Test for Linear Thermal Expansion of Fired Whiteware Products by the Didatometer Method.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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3. REQUIREMENTS

3.1 Material.

3.1.1 Porcelain. The casseroles and casserole covers shall be a high quality vitreous ceramic whiteware having an average thermal coefficient of linear expansion not greater than 3.6×10^{-6} centimeter (cm)/cm—°Centigrade (°C) between 20° and 200°C when tested in accordance with 4.4.1.

3.1.1.1 Thermal shock resistance. No cracking shall occur when casseroles and casserole covers are tested as specified in 4.4.2.

3.1.1.2 Glaze. The glaze shall offer a high degree of inertness and shall have a thermal coefficient of linear expansion approximately equal to that of the porcelain body. Type I, class 1 casseroles shall be glazed inside and outside with the exception of the rim and outside bottom. Type I, class 2 casseroles shall be glazed inside and outside except for the outside bottom. Type II casseroles shall be glazed inside and outside with exception of the outside bottom and the threaded projection for wooden handle. Casserole covers shall be glazed inside and outside.

3.1.1.2.1 Glaze adhesion. Permanent adhesion of the glaze to another glazed porcelain surface of the same material shall not occur below 1150°C when tested as specified in 4.4.3.

3.1.1.2.1.1 Permanent adhesion. Adhesion shall be considered permanent when, after cooling to room temperature, the touching surfaces are annealed adhering to and becoming part of the other surface.

3.1.1.2.2 Resistance to chemical attack. Resistance to chemical attack shall be measured by loss of weight by boiling in a solution of sodium hydroxide. Weight loss shall be not more than 0.50 percent when tested as specified in 4.4.4.

3.1.1.2.3 Solubility. No watermark shall remain on the inner surface of the casserole and no pink color shall develop in the test solution when tested as specified in 4.4.5.

3.1.2 Handles.

3.1.2.1 Type I. Handles for Type I casseroles shall be fabricated from porcelain described in 3.1.1 and shall be molded integrally with the cup.

3.1.2.2 Type II. Handles for type II casseroles shall be fabricated from hard wood suitable for the purpose, turned to the shape of a handle, rounded at one end with the other end threaded to fit securely the threaded projection from the cup. Handles may also be secured by a pin through the cup projection to engage the shank of the handle or securely attached with a heat-resistant thermosetting adhesive or epoxy cement. Handles shall have a protective finish.

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3.2 Design:

3.2.1 Shape.

3.2.1.1 Type I. Casseroles shall have sloping and convex sides extending up to the rim. Casseroles shall have a pour-out spout located 90 degrees to the left of the handle when the handle is towards the viewer. The spout shall pour evenly when tested as specified in 4.4.6. The handle of the casserole shall be flat with a hole bored in the end to permit suspension of the casserole by hanging. The bottom of the casserole shall be flattened for stability.

3.2.1.2 Type II. Casseroles shall have relatively straight sides rounded at the junction with the flat bottom. Bottom diameter shall be only slightly less than top diameter. Casseroles shall have a pour-out spout located 90 degrees to the left of the handle when the handle is towards the viewer. The spout shall pour evenly when tested as specified in 4.4.6.

3.2.1.3 Covers. Covers for Type II casseroles shall have a diameter slightly larger than the top of the cup with a projecting edge to fit over the cup. In the center of the top of the cover, there shall be a molded knob to serve as a handle.

3.2.2 Dimensions and capacities. Dimensions and capacities shall be as shown in table I.

TABLE I. Dimensions and capacities of casseroles.

Type	Class	Capacity ** (ml)	Dimensions	
			Top diameter * (mm)	Height * (mm)
I	1	30	50	25
I	1	140	85	45
I	1	210	95	50
I	1	340	110	62
I	1	500	130	73
I	1	750	135	80
I	2	300	108	57
II	--	150	90	45

* Tolerance ± 5 mm
** Tolerance shall not exceed 15%

3.3 Identification marking. Each casserole and casserole cover shall be permanently and legibly marked with the name or trademark of the manufacturer. Markings shall be legible after shock and chemical tests have been performed.

3.4 Workmanship. Workmanship shall be first class throughout. Casseroles shall be free of defects listed in table IV and any other defects which detract from their appearance or impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III. Acceptance number shall be zero.

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TABLE III. Sampling for tests.

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level *
Material	Coefficient of expansion	3.1.1	4.4.1	*
Casserole	Thermal shock resistance	3.1.1.1	4.4.2	S-2
	Glaze adhesion	3.1.1.2.1	4.4.3	S-2
	Resistance to chemical attack	3.1.1.2.2	4.4.4	S-2
	Solubility	3.1.1.2.3	4.4.5	S-2
	Pour test	3.2.1	4.4.6	S-2

* Sufficient sample shall be taken to perform test.

4.3 Examination. The casseroles shall be examined to determine compliance with all requirements contained in this specification. Definitions of terms shall be as indicated in ASTM C242. Examination shall be in accordance with but shall not be limited to the defects listed in table IV.

TABLE IV. Classification of defects.

Categories	Defects
Major	
101	Casserole not free of cracks.
102	Casserole not free of chips.
103	Casserole not free of deformation.
104	Casserole not free of blistering.
105	Glaze not continuous.
106	Component part missing.
107	Component not free from unspecified protrusions.
108	Surface not smooth (except rim).
109	Handle does not adhere.
Minor	
201	Casserole not free of crazing.
202	Casserole not free of shivering.
203	Casserole not free of egg shelling.
204	Casserole not free of orange peel.
205	Casserole not free of crawling.
206	Casserole not free of foreign matter.
207	Identification marking not complete, correct or legible.

4.3.1 Dimensional examination. The casseroles shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be considered as a defect.

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4.3.2 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking comply with the requirements of Section 5. Defects shall be scored as specified in table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

TABLE V. Examination of preparation for delivery.

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and Packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, tape improperly applied or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.

4.4 Tests.

4.4.1 Coefficient of expansion. Porcelain shall be tested for linear coefficient of expansion in accordance with ASTM C327 or C372.

4.4.2 Thermal shock resistance. The casseroles (and covers) shall be heated to $160^{\circ} \pm 2^{\circ}C$ in an oven for 30 minutes, removed, and immediately plunged into water at a temperature of $9^{\circ} \pm 1^{\circ}C$.

4.4.3 Glaze adhesion. Place 2 casseroles (and 2 covers) so that the glazed surfaces are in contact with each other (casserole to casserole and cover to cover) and heat to $1150^{\circ}C$ in an electric muffle-type furnace for 30 minutes. Allow to cool by radiation to room temperature.

4.4.4 Resistance to chemical attack. Weighed casserole (and cover) shall be boiled in 10 percent solution of sodium hydroxide for 8 hours, rinsed, dried and reweighed. Weight loss shall be calculated as follows:

$$L = \frac{100 \quad (B-A)}{B}$$

where:

L = percent weight loss
B = weight, before boiling, in grams
A = weight, after boiling, in grams

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4.4.5 Solubility. Each casserole (and cover), taken for test, shall be cleaned and rinsed three times with water, using a new quantity of water for each rinsing. The casserole (and cover) shall be air-dried in an inverted position for 1 minute after each rinsing. After the final drying operation, each casserole shall be half-filled with water at 60°C containing 0.2 percent phenolphthalein solution (prepared from 1 gram of phenolphthalein in 100 ml of alcohol) and autoclaved under 15 pounds steam pressure for 30 minutes. Casserole covers shall be immersed individually in an upside down position in water containing 0.2 percent phenolphthalein solution in a beaker of suitable size. The water used in this solubility test shall have just been boiled 10 minutes and allowed to cool to 60°C, prior to the addition of the phenolphthalein solution.

Note: Glassware used in this test shall not affect test solution.

4.4.6 Pour test. Fill the casserole to indicated capacity with water. Pour approximately one-fourth of the water from the casserole at such a rate that the water flows in a continuous stream and falls free from the lip of the casserole. If the stream of water clings to the outside of the casserole and flows off the bottom, it shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Unit of issue. One casserole as specified constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each casserole shall be packaged in a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, type CF, class domestic, except that commercial colors shall be acceptable. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure shall be adequate to prevent accidental opening.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Forty-eight unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the box specification.

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5.2.2 Level B.

5.2.2.1 Exterior container. Forty-eight unit packages shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Type I, class 1 and Type II casseroles are intended for general laboratory use; Type I, class 2 is intended for silica determination.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Applicable levels of packaging and packing (see 5.1 and 5.2).
d. Type, class and capacity required.

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